VAISHNAVI M

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EDUCATION

B.E. in Computer Science & Engineering | Bangalore Institute of Technology,Bangalore |*2023–2027*
CGPA: 8.9 | Relevant Coursework: Data Structures, DBMS, OS, Computer Networks, OOP, Cloud Computing

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML/CSS
Frameworks: Flask, Node.js, React (basics)
Databases: MySQL, phpMyAdmin
Tools: Git, GitHub, XAMPP, Streamlit, AWS (EC2, Lambda, DynamoDB, API Gateway), Docker
Concepts: OOP, DBMS, OS, CN, DSA, REST APIs, Machine Learning, Blockchain (basics), Federated Learning
Security: Cyber Threat Intelligence, Intrusion Detection, XAI (SHAP/LIME), ZK Proofs (theoretical)

PROJECTS

AI-Powered Decentralized Cyber Threat Intelligence Network | Final Year Project | *2024 – 2025*

- Built a decentralized CTI sharing platform combining blockchain, federated learning, and explainable AI (SHAP/LIME).
- Implemented Zero-Knowledge Proofs for privacy-preserving threat sharing; differentiated from tools like VirusTotal and MISP with ~40% reduction in false positive rate vs baseline centralized model.
- Designed a token economy to incentivize cross-organization intelligence contribution; system validated across 3 federated nodes with <2s consensus latency.
- Tech stack: Python, Blockchain, Federated Learning, XAI (SHAP/LIME), ZK Proofs

EdCare – Elder Care Health Support System | Backend Project | *2025*

- Built and deployed a production-grade elder care REST API handling symptom triage, medicine reminders, activity tracking, and caregiver monitoring; average API response time <350ms on Railway cloud.
- Rule-based triage engine (WHO/ICMR guidelines) covers 50+ symptoms across 4 severity levels with synonym mapping; medical history-aware logic reduces missed high-risk cases by upgrading severity for flagged conditions.
- Implemented JWT authentication with bcrypt, daily health scoring algorithm, and caregiver dashboard; system exposes 20+ REST endpoints with full audit trail.
- Deployed on Railway cloud with MySQL; followed Flask Blueprint architecture for modular, scalable code.
- Tech stack: Python, Flask, MySQL, JWT, REST API, Railway (Cloud Deployment)

PhishGuard – AI-Powered Phishing Detection Browser Extension | Cybersecurity & Machine Learning | 2026

- Built an AI-powered browser extension that detects phishing websites in real time using a **Random Forest** machine learning model trained on phishing URL datasets; provides instant security alerts before users interact with malicious websites.

- Developed a feature extraction pipeline to analyze **URL- and domain-based characteristics** for phishing classification, enabling accurate identification of suspicious websites through REST API inference.
- Integrated the trained model with a **Flask REST API** and Chrome Extension to deliver low-latency predictions, creating a seamless real-time phishing detection workflow.
- Implemented a modular architecture separating the browser extension, backend prediction service, and machine learning model for improved scalability and maintainability.
- **Tech Stack:** Python, Flask, Scikit-learn, Random Forest, JavaScript, HTML, CSS, Chrome Extension API, REST API

Blood Bank Availability Management System | DBMS Project | *2023*

- Designed and deployed a full-stack web app for real-time blood availability tracking with admin and user access roles; supports 6 blood groups with instant availability lookup (<200ms query response).
- Features: donor/patient management, blood request tracking, inventory updates; handled 3 concurrent user roles with role-based access control.
- Tech stack: Python (Flask), MySQL, XAMPP, phpMyAdmin, HTML/CSS

Network Intrusion Detection System (NIDS) | Neural Networks Mini-Project | *2024*

- Developed an ANN-based NIDS on NSL-KDD dataset achieving 98.2% training accuracy and 94.7% test accuracy with <0.3s average inference latency per sample.
- Model achieved F1-score of 0.93 across attack categories (DoS, Probe, R2L, U2R); deployed as an interactive web app using Streamlit.
- Tech stack: Python, TensorFlow/Keras, Streamlit, Pandas, Scikit-learn

GrowDesk – AI-Powered Micro-Business Toolkit | Side Project | *2024*

- Built a standalone MVP with Claude API integration featuring invoicing, task management, and AI-driven idea validation; average AI response time ~1.2s per query.
- Designed with freemium monetization model targeting micro-businesses and solopreneurs; supports invoice generation, task CRUD, and real-time AI feedback in a single-page app.
- Tech stack: HTML, CSS, JavaScript, Claude API (Anthropic)

INTERNSHIP

Cyber Security Intern | Threat Prism (via 1stop.ai) | *Sep 2025 – Nov 2025*

- Completed an Industrial Program in Cyber Security, working on 3 hands-on security projects under mentor guidance.
- Phishing Awareness Simulation: Designed and executed simulated phishing campaigns to assess and improve user awareness.
- Android Application Penetration Testing: Performed static and dynamic analysis to identify vulnerabilities in Android apps.
- Digital Forensic Analysis with Autopsy: Conducted forensic investigations on disk images using Autopsy to recover and analyze digital evidence.

CERTIFICATIONS & COURSES

- Cyber Security Industrial Program – Threat Prism / 1stop.ai (Sep – Nov 2025)
- Anthropic Academy – Claude API Fundamentals, MCP, Agent Skills (in progress)
- AWS Cloud Practitioner Essentials (EC2, Lambda, DynamoDB, API Gateway, Docker)
- TryHackMe – Cybersecurity labs and CTF challenges
- LeetCode / GeeksforGeeks – DSA practice (Arrays, Trees, Graphs, DP, Sorting)

ACHIEVEMENTS & ACTIVITIES

- Active Member, eSwaccha Club – Contributed to e-waste awareness and responsible electronics disposal initiatives on campus.
- Fest Organizing Committee Member, MANTHAN 2025 – Part of the organizing team for the college annual fest, managing event coordination and logistics.
- Actively preparing for campus placements: DSA, DBMS, OS, OOP, CN, and Aptitude (evening study plan, 2-3 hrs/day).